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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 11/12/2025 |
| **PI Name:** | Urbaniak, Camilla Ph.D. | | |
| **Project Title:** | Microgravity Analogs as Proxies for Spaceflight to Validate Biofilm Formation and the Exchange of Genes Between Organisms | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Microbiology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | camilla.urbaniak@jpl.nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | INDUSTRY | **Phone:** | 626-840-6992 |
| **Organization Name:** | ZIN Technologies, Inc | | |
| **PI Address 1:** | 4800 Oak Grove Dr | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Pasadena | **State:** | CA |
| **Zip Code:** | 91109-8001 | **Congressional District:** | 27 |
| **Comments:** | NOTE: PI had NASA Postdoctoral Program Fellowship at NASA Jet Propulsion Laboratory until 2019; now with Zin Technologies, Inc., Middleburg, OH, but remains physically at NASA JPL address. | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | 2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research |
| **Start Date:** | 01/04/2021 | **End Date:** | 01/03/2027 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Jones, Harry | **Contact Phone:** | 650.604.5518 |
| **Contact Email:** | harry.jones@nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: End date changed to 01/03/2027 per F. Hernandez/ARC (Ed., 1/22/26).<br>NOTE: End date changed to 01/03/2026 per F. Hernandez/ARC (Ed., 12/20/24).<br>NOTE: End date changed to 01/03/2025 per F. Hernandez/ARC (Ed., 6/18/24). | | |
| **Key Personnel Changes/Previous PI:** | Per the Principal Investigator, a new Co-Investigator has been added to the research team. Wayne Schubert from the Jet Propulsion Laboratory has been added to the team. Dr. Max Coleman has left the investigation (Ed., 11/19/25). | | |
| **COI Name (Institution):** | Schubert, Wayne ( NASA Jet Propulsion Laboratory ) | | |
| **Grant/Contract No.:** | 80NSSC21K0320 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Microbial adaptation to space is an important area to study as it provides information on how microbes on spacecraft and the microbiome of astronauts will change when exposed to the space environment and how that may impact spacecraft integrity and function, and the health of astronauts. However, due to time and cost constraints of conducting studies on the International Space Station (ISS), scientists are not able to perform as many insightful space biology studies as they would like. For this reason, scientists use microgravity simulators to conduct space biology studies, but on Earth. However, there are very limited studies testing how these simulators compare to the "real" thing and the jury is still out on whether microgravity simulators are actually comparable to experiments performed on the ISS. For this reason, we wish to carry out a study that looks at how three simulated microgravity chambers, commonly used on the ground, compare to what would be observed on the ISS. This study would have a ground and a flight component and would look at two bacterial properties that are significant for short and long duration space missions--biofilm formation and the exchange of genetic material between bacteria. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | One of NASA’s goals is to minimize the health risks associated with extended spaceflight, so it is critical that methods for preventing and treating spaceflight-induced illnesses be developed before astronauts embark upon long-duration space missions. In order for this to happen, it is important to learn how bacteria that play a role in human health and disease are affected by spaceflight, such as biofilm formation and horizontal gene transfer. Comparing these responses to popular and frequently used simulated microgravity devices on Earth, and verifying whether the same outcomes can be achieved, will allow more researchers to partake in space related microbiology and human microbiome work, without the extensive limitations imposed by conducting experiments on the ISS. This ultimately diversifies the field and opens up opportunities for scientists from all backgrounds. In addition, understanding microbial adaptation in the new and unique environment of space may help guide future research questions and studies into microbial adaptations on Earth under changing conditions, such as climate change. |
| **Task Progress:** | All milestones and ground testing for the project have been completed. We are currently preparing for flight to the ISS. |
| **Bibliography Type:** | Description: (Last Updated: 11/01/2024) |